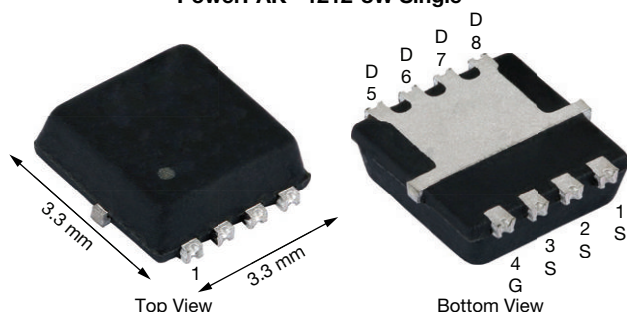


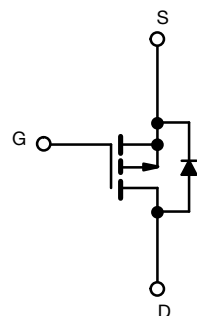
Automotive P-Channel 40 V (D-S) 175 °C MOSFET

PowerPAK® 1212-8W Single

Marking code: Q035

FEATURES

- TrenchFET® power MOSFET
- AEC-Q101 qualified
- 100 % R_g and UIS tested
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE
GRADE

RoHS
COMPLIANT
HALOGEN
FREE


P-Channel MOSFET

PRODUCT SUMMARY	
V _{DS} (V)	-40
R _{DS(on)} (Ω) at V _{GS} = -10 V	0.0273
R _{DS(on)} (Ω) at V _{GS} = -4.5 V	0.0380
I _D (A)	-16
Configuration	Single
Package	PowerPAK 1212-8W

ABSOLUTE MAXIMUM RATINGS (T _C = 25 °C, unless otherwise noted)				
PARAMETER		SYMBOL	LIMIT	UNIT
Drain-source voltage		V _{DS}	-40	V
Gate-source voltage		V _{GS}	± 20	
Continuous drain current ^a	T _C = 25 °C	I _D	-16	A
	T _C = 125 °C		-16	
Continuous source current (diode conduction) ^a		I _S	-16	
Pulsed drain current ^b		I _{DM}	-64	
Single pulse avalanche current	L = 0.1 mH	I _{AS}	-19	
Single pulse avalanche energy		E _{AS}	18	mJ
Maximum power dissipation	T _C = 25 °C	P _D	39.5	W
	T _C = 125 °C		13.2	
Operating junction and storage temperature range		T _J , T _{stg}	-55 to +175	°C
Soldering recommendations (peak temperature) ^{d, e}			260	

THERMAL RESISTANCE RATINGS				
PARAMETER		SYMBOL	LIMIT	UNIT
Junction-to-ambient	PCB mount ^c	R _{thJA}	81	°C/W
Junction-to-case (drain)		R _{thJC}	3.8	

Notes

- Package limited
- Pulse test; pulse width ≤ 300 μs, duty cycle ≤ 2 %
- When mounted on 1" square PCB (FR4 material)
- See solder profile (www.vishay.com/doc?73257). The PowerPAK 1212-8W is a leadless package. The end of the lead terminal is exposed copper (not plated) as a result of the singulation process in manufacturing. A solder fillet at the exposed copper tip cannot be guaranteed and is not required to ensure adequate bottom side solder interconnection
- Rework conditions: manual soldering with a soldering iron is not recommended for leadless components



SPECIFICATIONS (T _C = 25 °C, unless otherwise noted)							
PARAMETER	SYMBOL	TEST CONDITIONS		MIN.	TYP.	MAX.	UNIT
Static							
Drain-source breakdown voltage	V _{DS}	V _{GS} = 0 V, I _D = -250 μA		-40	-	-	V
Gate-source threshold voltage	V _{GS(th)}	V _{DS} = V _{GS} , I _D = -250 μA		-1.5	-2.0	-2.5	
Gate-source leakage	I _{GSS}	V _{DS} = 0 V, V _{GS} = ± 20 V		-	-	± 100	nA
Zero gate voltage drain current	I _{DSS}	V _{GS} = 0 V	V _{DS} = -40 V	-	-	-1	μA
		V _{GS} = 0 V	V _{DS} = -40 V, T _J = 125 °C	-	-	-50	
		V _{GS} = 0 V	V _{DS} = -40 V, T _J = 175 °C	-	-	-150	
On-state drain current ^a	I _{D(on)}	V _{GS} = -10 V	V _{DS} ≤ -5 V	-20	-	-	A
Drain-source on-state resistance ^b	R _{DS(on)}	V _{GS} = -10 V	I _D = -8 A	-	0.0210	0.0273	Ω
		V _{GS} = -10 V	I _D = -8 A, T _J = 125 °C	-	-	0.0405	
		V _{GS} = -10 V	I _D = -8 A, T _J = 175 °C	-	-	0.0480	
		V _{GS} = -4.5 V	I _D = -6 A	-	0.0290	0.0380	
Forward transconductance ^b	g _{fs}	V _{DS} = -15 V, I _D = -7 A		-	23	-	S
Dynamic ^b							
Input capacitance	C _{iss}	V _{GS} = 0 V	V _{DS} = -25 V, f = 1 MHz	-	2455	3191	pF
Output capacitance	C _{oss}			-	157	205	
Reverse transfer capacitance	C _{rss}			-	228	187	
Total gate charge ^c	Q _g	V _{GS} = -10 V	V _{DS} = -20 V, I _D = -2.5 A	-	38	50	nC
Gate-source charge ^c	Q _{gs}			-	6	8	
Gate-drain charge ^c	Q _{gd}			-	7	10	
Gate resistance	R _g	f = 1 MHz		2.5	4.2	6.7	Ω
Turn-on delay time ^c	t _{d(on)}	V _{DD} = -20 V, R _L = 14.3 Ω I _D ≅ -1.4 A, V _{GEN} = -10 V, R _g = 1 Ω		-	10.5	13.7	ns
Rise time ^c	t _r			-	3.0	3.9	
Turn-off delay time ^c	t _{d(off)}			-	39.6	51.5	
Fall time ^c	t _f			-	6.4	8.4	
Source-Drain Diode Ratings and Characteristics ^b							
Pulsed current ^a	I _{SM}			-	-	-64	A
Forward voltage	V _{SD}	I _F = -8 A, V _{GS} = 0 V		-	-0.8	-1.2	V

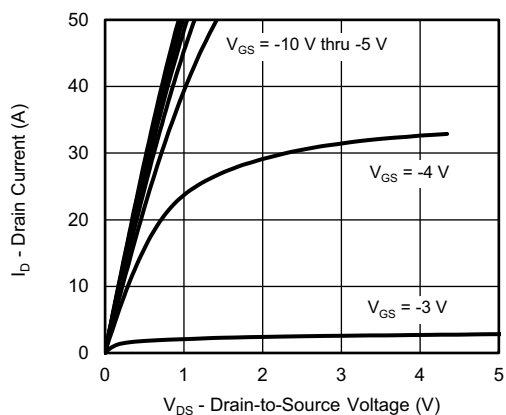
Notes

- a. Pulse test; pulse width $\leq 300\text{ }\mu\text{s}$, duty cycle $\leq 2\%$
b. Guaranteed by design, not subject to production testing
c. Independent of operating temperature

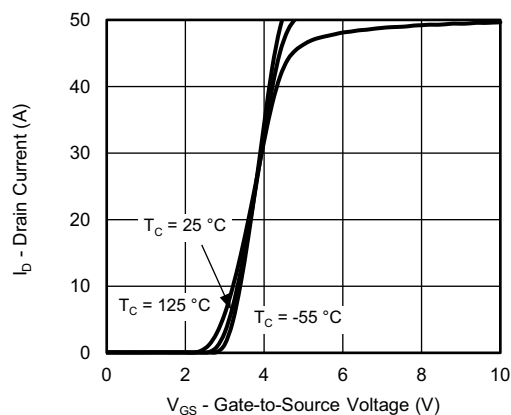
Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.



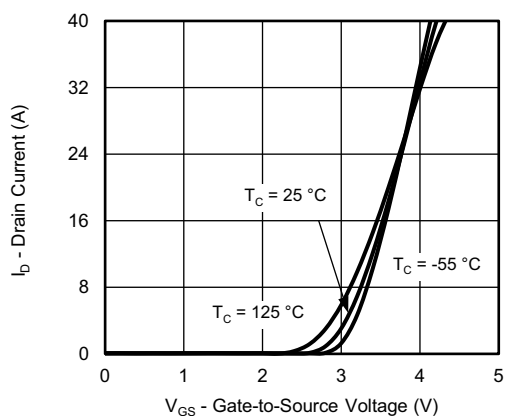
TYPICAL CHARACTERISTICS ($T_A = 25\text{ }^{\circ}\text{C}$, unless otherwise noted)



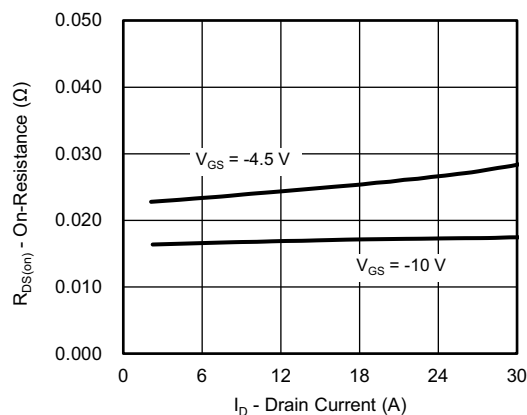
Output Characteristics



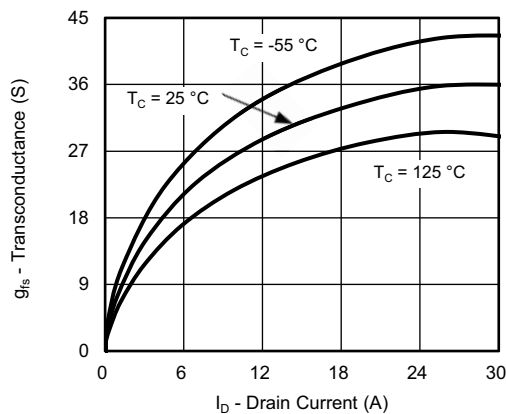
Transfer Characteristics



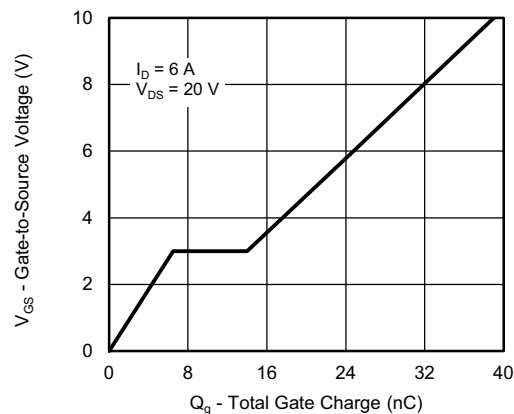
Transfer Characteristics



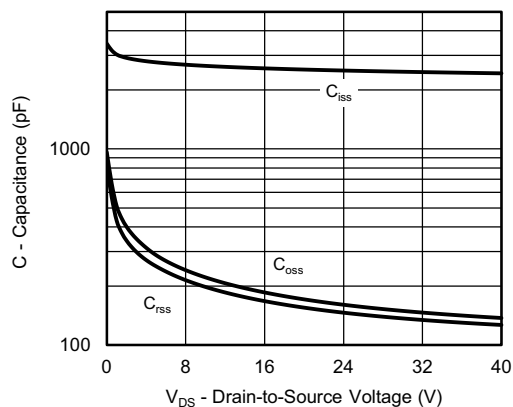
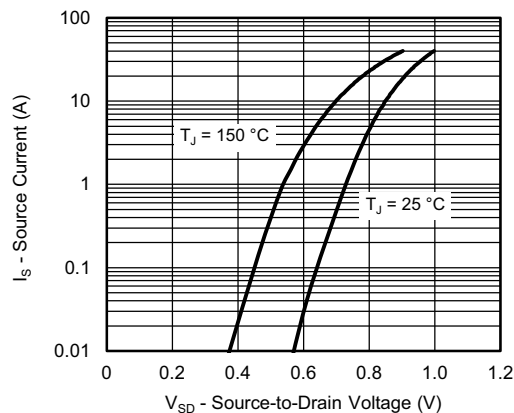
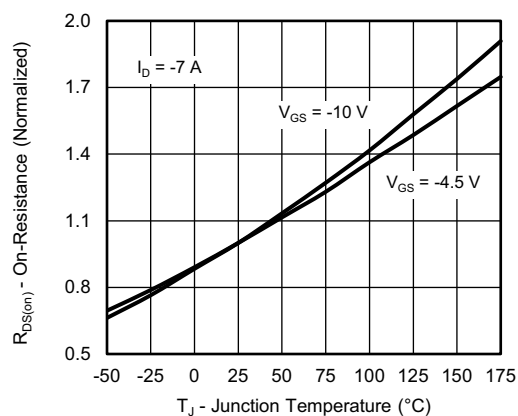
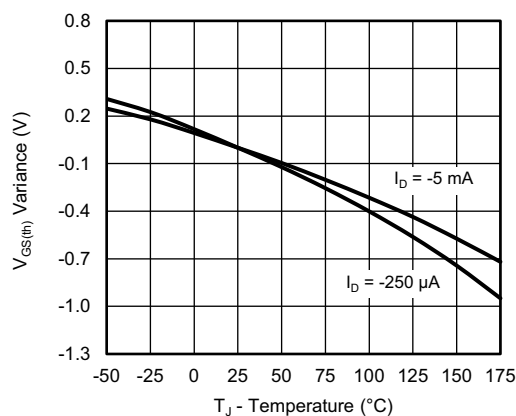
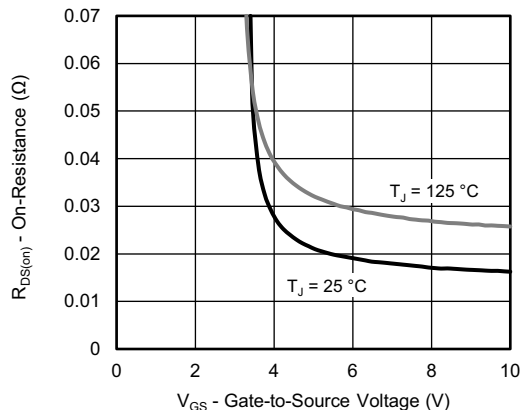
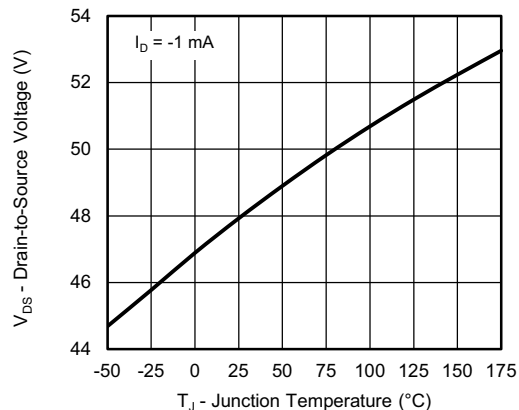
On-Resistance vs. Drain Current



Transconductance

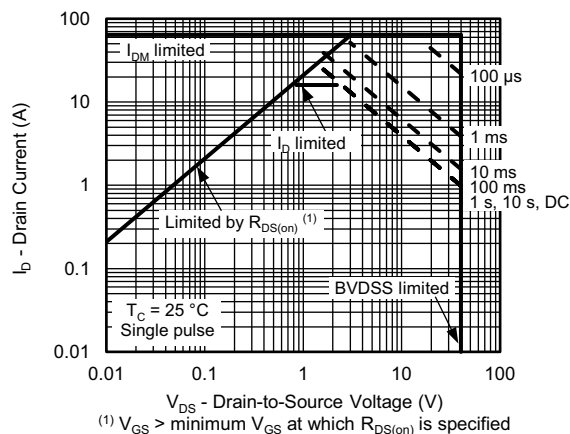


Capacitance

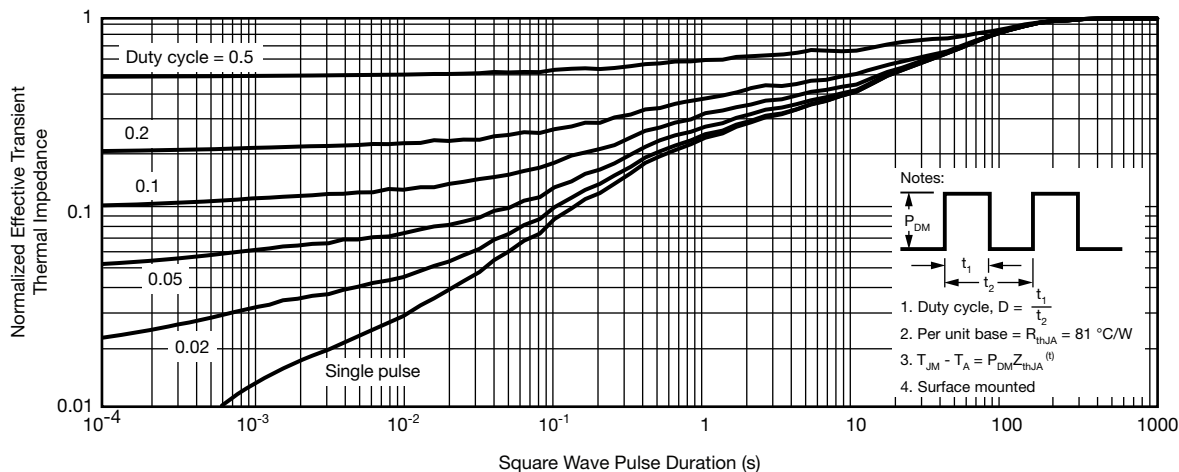
TYPICAL CHARACTERISTICS ($T_A = 25\text{ }^{\circ}\text{C}$, unless otherwise noted)

Capacitance

Source Drain Diode Forward Voltage

On-Resistance vs. Junction Temperature

Threshold Voltage

On-Resistance vs. Gate-to-Source Voltage

Drain Source Breakdown vs. Junction Temperature



THERMAL RATINGS ($T_A = 25\text{ }^{\circ}\text{C}$, unless otherwise noted)



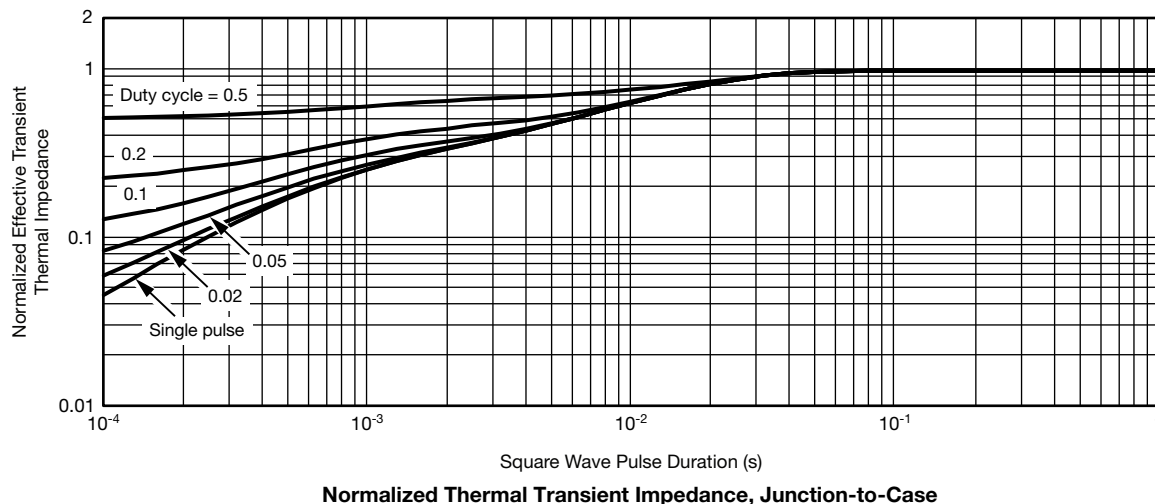
Safe Operating Area



Normalized Thermal Transient Impedance, Junction-to-Ambient



THERMAL RATINGS ($T_A = 25\text{ }^{\circ}\text{C}$, unless otherwise noted)

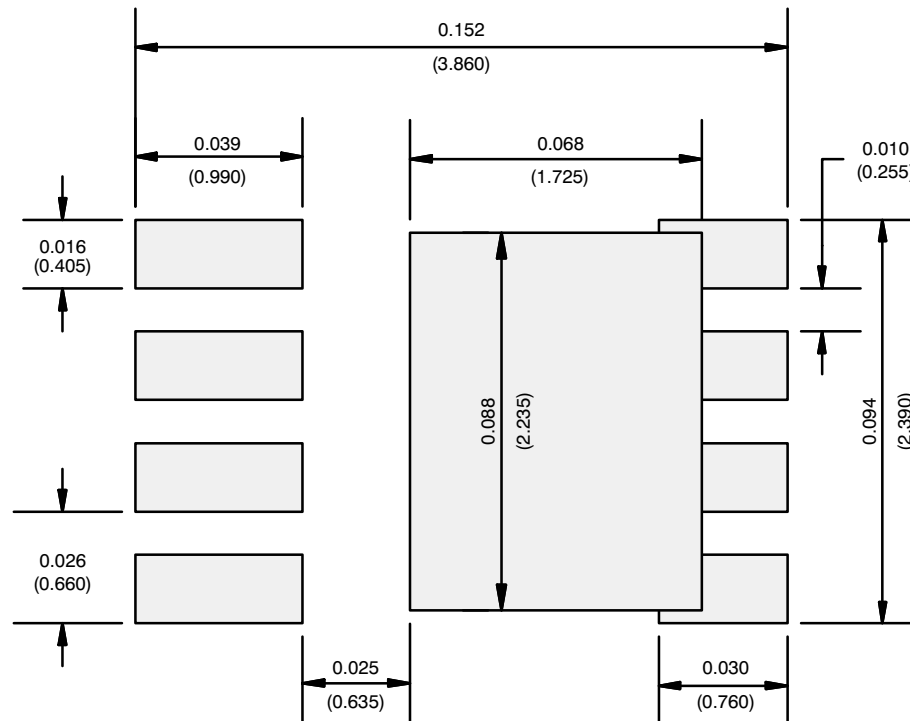


Note

- The characteristics shown in the two graphs
 - Normalized Transient Thermal Impedance Junction-to-Ambient ($25\text{ }^{\circ}\text{C}$)
 - Normalized Transient Thermal Impedance Junction-to-Case ($25\text{ }^{\circ}\text{C}$)are given for general guidelines only to enable the user to get a “ball park” indication of part capabilities. The data are extracted from single pulse transient thermal impedance characteristics which are developed from empirical measurements. The latter is valid for the part mounted on printed circuit board - FR4, size 1" x 1" x 0.062", double sided with 2 oz. copper, 100 % on both sides. The part capabilities can widely vary depending on actual application parameters and operating conditions

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RECOMMENDED MINIMUM PADS FOR PowerPAK® 1212-8 Single



Recommended Minimum Pads
Dimensions in Inches/(mm)

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